Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code: 3110016 **Date: 08-01-2019**

Subject Name: basic electronics

Time: 10:30 am to 01:00 pm **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	Explain V-I characteristic of tunnel diode.	**03**
	(b)	What is zener breakdown? What is avalanche breakdown? Compare both the type of breakdown.	**04**
	(c)	Write a short note: V-I characteristic of P-N junction diode.	**07**
Q.2	**(a)**	Design and explain basic NAND gate using DTL logic.	**03**
	(b)	Explain following gate using their truth table, logic symbol and equation. Ex-NOR, NAND, NO	**04**
	(c)	Draw and Explain bridge rectifier. Explain advantage and disadvantage of bridge rectifier over full wave rectifier.	**07**
		OR	
	(c)	Write a short note: Biased clipper circuit.	**07**
Q.3	**(a)**	Derive the relation between current gain **α** and **β**	**03**
	(b)	What is DC load line? Explain with necessary diagram.	**04**
	(c)	Draw and explain input and output characteristic of transistor in common emitter configuration.	**07**
		OR	
Q.3	**(a)**	What is stability factor? Explain.	**03**
	(b)	Give comparison between CE, CB and CC configuration of transistor.	**04**
	(c)	What are the different method for biasing the transistor. Explain any two method with necessary circuit diagram.	**07**
Q.4	**(a)**	Why biasing circuits are required?	**03**
	(b)	Explain why NAND and NOR gate are called universal gate?	**04**
	(c)	Explain application of transistor as a switch.	**07**
		OR	
Q.4	**(a)**	List out the salient feature of emitter follower.	**03**
	(b)	Explain various properties of CB amplifier.	**04**
	(c)	Draw and explain the transistor a.c. equivalent circuit.	**07**
Q.5	**(a)**	Give comparison of BJT and JFET.	**03**
	(b)	Draw and explain the self bias circuit of FET.	**04**
	(c)	Draw and explain various characteristic of JFET	**07**
		OR	
Q.5	**(a)**	What are the advantage of N-Channel MOSFET over P-Channel MOSFET.	**03**
	(b)	Explain the application of FET as a buffer amplifier.	**04**
	(c)	Write a short note: E-Type MOSFET	**07**

Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I &II (NEW) EXAMINATION – SUMMER-2019

Subject Code: 3110016 **Date: 07/06/2019**

Subject Name: Basic Electronics

Time: 10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	Differentiate between insulator, conductor and semiconductor	**03**
	(b)	Explain forward bias PN junction diode with diagram	**04**
	(c)	Explain full wave bridge rectifier with neat diagram	**07**
Q.2	**(a)**	Explain LED diode	**03**
	(b)	State different types of diodes. Describe process of testing diode with multi meter.	**04**
	(c)	What is break down diode?? Explain working of zener break down and avalanche break down	**07**
		OR	
	(c)	Why biasing is important in transistor? Explain voltage divider bias with diagram.	**07**
Q.3	**(a)**	What is use of coupling and bypass capacitor?	**03**
	(b)	Explain PIN photo diode	**04**
	(c)	Draw the circuit of transistor in CE configuration. Sketch the output characteristics and explain active, saturation and cutoff regions	**07**
		OR	
Q.3	**(a)**	What is varactor diode? How capacitance of a diode varies with reverse voltage?	**03**
	(b)	Explain AC loadline with respect to BJT	**04**
	(c)	Compare CE, CB and CC configuration with respect to different transistor characteristics	**07**
Q.4	**(a)**	What is FET? State important features of FET.	**03**
	(b)	Compare BJT and FET	**04**
	(c)	Write short note on MOSFET.	**07**
		OR	
Q.4	**(a)**	Explain clipping circuit	**03**
	(b)	Explain (i) Unipolar device (ii) Transconductance	**04**
	(c)	Write shortnote on JFET	**07**
Q.5	**(a)**	Draw the symbol of NPN and PNP transistor. What is use of transistor?	**03**
	(b)	Among TTL and CMOS digital logic family which one is better and why?	**04**
	(c)	Draw symbol and explain truth table of all basic logic gates	**07**
		OR	
Q.5	**(a)**	State advantage of transistor	**03**
	(b)	Explain (i)universal gate (ii) EX-OR logic gate	**04**
	(c)	Give comparison between different types of digital logic families	**07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019

Subject Code: 3110016 **Date: 06/01/2020**

Subject Name: Basic Electronics

Time: 10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	Draw the circuit diagram of Half wave rectifier.	**03**
	(b)	Explain the bridge rectifier with diagrams.	**04**
	(c)	Determine the Vo for the network shown in figure 1	**07**

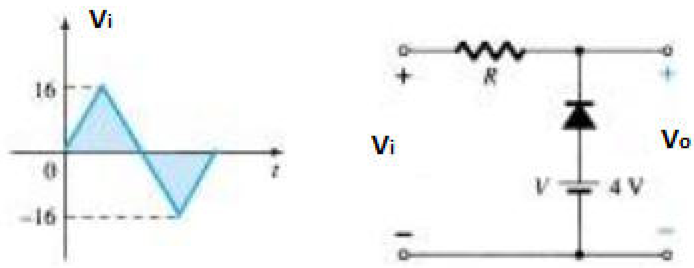

Figure 1

Q.2	**(a)**	Explain Varactor diode and varistor.	**03**
	(b)	Why Zener diode can be used as voltage regulator? Explain Zener as voltage regulator with necessary diagram	**04**
	(c)	Compare the logic families and explain any one of them.	**07**
		OR	
	(c)	Explain Ex-OR and Ex- NOR gate with truth table and construct OR gate using diodes.	**07**
Q.3	**(a)**	Explain about DC load line and Bias point of transistor	**03**
	(b)	Explain the working of PIN Diode.	**04**
	(c)	Briefly explain the h-parameters and draw h-parameter based equivalent circuit for CE transistor and derive equation for input impedance, output impedance and voltage gain.	**07**
		OR	
Q.3	**(a)**	Write truth table of AND, NAND and NOR gates.	**03**
	(b)	Explain the selection of a Q point for a transistor bias circuit and discuss the limitations on the output voltage swing.	**04**
	(c)	Explain the difference between clipping and clamping circuit. A positive voltage clamping circuit and a positive voltage clipping circuit each have ±12 V square Wave input. Sketch the output waveform for each circuit.	**07**
Q.4	**(a)**	Draw voltage multiplier circuit.	**03**
	(b)	Explain Transconductance and switching in FET.	**04**

(c) Explain the Depletion region and drain characteristics of n channel JFET. **07**

OR

Q.4 **(a)** Discuss about VI characteristic of Ideal Diode. **03**

(b) Explain FET as an Amplifier. **04**

(c) Determine the voltage Vo for the network of Figure 2. Give explanation for your answer. **07**

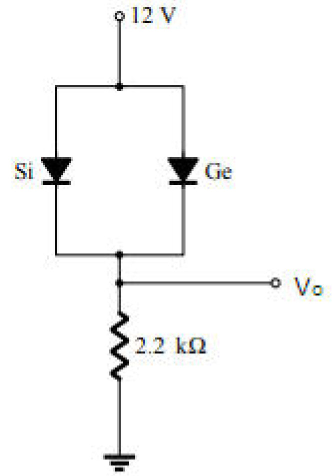

Figure 2

Q.5 **(a)** Explain the working of Transistor as Switch **03**

(b) Write a short note on E MOSFET as an Amplifier. **04**

(c) Design a series noise clipping circuit which rectify the noise signal with amplitude lower than $\pm V_F$. **07**

OR

Q.5 **(a)** Explain the AC load line of transistor. **03**

(b) Draw and explain seven segment display. **04**

(c) Compare BJT with FET and explain D MOSFET. **07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II(NEW)EXAMINATION – SUMMER 2022

Subject Code:3110016 **Date:24-08-2022**

Subject Name:Basic Electronics

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	What is a diode? Write its types and applications.	**03**
	(b)	Explain the diode V-I characteristics of ideal and practical PN junction semiconductor diode.	**04**
	(c)	Enumerate the different types of clipping circuits with their different names and input-output waveforms.	**07**
Q.2	**(a)**	Why are junction transistors called bipolar devices?	**03**
	(b)	The metal lead of the p-side of a p–n diode is soldered to the metal lead of the p-side of another p–n junction diode. Will the structure form an n–p–n transistor? If not, why?	**04**
	(c)	Sketch the circuit of the common collector mode of BJT and its output characteristics. Derive the expression for the collector current and gain.	**07**
		OR	
	(c)	Draw the fixed-biased circuit by considering an n–p–n transistor in the CE mode. Derive the expressions for stability factors. What are the functions of the coupling capacitors?	**07**
Q.3	**(a)**	Write a short note on the optocoupler device?	**03**
	(b)	Explain the sixteen segment display and its applications with the necessary circuit diagram.	**04**
	(c)	Draw the approximate hybrid model for any transistor configuration at low frequencies. Show that only h_{ie} and h_{fe} are essential in the model. Is the approximation justified?	**07**
		OR	
Q.3	**(a)**	Explain the varactor diode.	**03**
	(b)	Explain the contraction of the solar cell with its operational principle.	**04**
	(c)	What is self-bias? Draw the circuit showing self-bias of an n–p–n transistor in the CE mode. Explain physically how the self-bias improves stability.	**07**
Q.4	**(a)**	What is MOSFET device? Draw its construction diagram.	**03**
	(b)	Write short notes on the following : (i) Advantages of JFET (ii) Difference between MOSFET and JFET	**04**
	(c)	Compare the different characteristics of the following semiconductor devices: bipolar junction transistor, field-effect transistor.	**07**
		OR	
Q.4	**(a)**	How will you determine the drain characteristics of JFET? What do they indicate?	**03**
	(b)	Explain the common drain configuration for a JFET.	**04**
	(c)	Explain the JFET parameters and establish the relationship between them	**07**

Q.5	**(a)**	What is the thermal runaway in transistors, and how can it be avoided?	**03**
	(b)	What is an Early effect, and how can it account for the CB input characteristics?.	**04**
	(c)	What do you mean by the logic gate and its types? Explain the universal logic gates.	**07**

OR

Q.5	**(a)**	What is the ac load line in the transistor? Write its significance.	**03**
	(b)	The value of alpha increases with the increasing reverse-bias voltage of the collector junction. Why?	**04**
	(c)	Explain the logic families and their types. Describe the characteristics of the same.	**07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I & II(NEW) EXAMINATION – WINTER 2022

Subject Code:3110016 Date:16-03-2023

Subject Name:Basic Electronics

Time:10:30 AM TO 01:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Define between insulator, conductor and semiconductor.	**03**
	(b)	What is Zener breakdown? What is avalanche breakdown? Compare both the type of breakdown.	**04**
	(c)	Explain full wave bridge rectifier with neat diagram.	**07**
Q.2	**(a)**	Design and explain basic NAND gate using DTL logic.	**03**
	(b)	Discuss different types of diodes. Describe process of testing diode with multi meter.	**04**
	(c)	Write a short note: V-I characteristic of P-N junction diode.	**07**
		OR	
	(c)	Why biasing is important in transistor? Explain voltage divider bias with diagram.	**07**
Q.3	**(a)**	Explain about DC load line of transistor.	**03**
	(b)	Explain PIN photo diode.	**04**
	(c)	Draw and explain input and output characteristic of transistor in common emitter configuration.	**07**
		OR	
Q.3	**(a)**	What is varactor diode? How capacitance of a diode varies with reverse voltage?	**03**
	(b)	Explain the selection of a Q point for a transistor bias circuit and discuss the limitations on the output voltage swing.	**04**
	(c)	Compare CB and CC configuration with respect to different transistor characteristics.	**07**
Q.4	**(a)**	Draw voltage multiplier circuit.	**03**
	(b)	Explain (i) Unipolar device (ii) Transconductance.	**04**
	(c)	Explain application of transistor as a switch.	**07**
		OR	
Q.4	**(a)**	Explain FET as an Amplifier.	**03**
	(b)	Explain why NAND and NOR gate are called universal gate?	**04**
	(c)	Explain the Depletion region and drain characteristics of n channel JFET.	**07**
Q.5	**(a)**	Among TTL and CMOS digital logic family which one is better and why?	**03**
	(b)	Draw and explain the self-bias circuit of FET.	**04**
	(c)	Write a short note: E-Type MOSFET.	**07**
		OR	
Q.5	**(a)**	Derive the relation between current gain α and β.	**03**
	(b)	What are the advantages of N-Channel MOSFET over P-Channel	**04**

MOSFET?

(c) Draw symbol and explain truth table of all basic logic gates. **07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II(NEW) EXAMINATION – SUMMER 2023

Subject Code:3110016 **Date:11-08-2023**

Subject Name:Basic Electronics

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Explain zener diode as a Voltage regulator.	**03**
	(b)	Compare Half wave rectifier and Full wave bridge rectifier for following parameters. (i) Average DC Voltage (ii) Ripple Factor (iii) PIV (iv) Rectifier Efficiency	**04**
	(c)	Illustrate the need of clipper circuit and explain Positive clipper and negative clipper with necessary diagram and waveforms.	**07**
Q.2	**(a)**	Describe the need of biasing the transistor.	**03**
	(b)	Draw the symbol of NPN & PNP transistor, write its different operating regions and also explain why it is called as “Bipolar Transistor”.	**04**
	(c)	Discuss Common Collector(CC) configuration of transistor and explain its input and output characteristics.	**07**
		OR	
	(c)	Discuss Common Base(CB) configuration of transistor and explain its input and output characteristics.	**07**
Q.3	**(a)**	Derive the relation between current gain α and β	**03**
	(b)	Describe the factors affecting the Stability of Q point.	**04**
	(c)	Explain the construction, working principle and application of phototransistor.	**07**
		OR	
Q.3	**(a)**	Discuss about leakage current in transistor.	**03**
	(b)	Write a short note on Thermal runaway.	**04**
	(c)	Describe the construction, the symbol, V-I characteristics and application of Tunnel diode.	**07**
Q.4	**(a)**	Discuss about transconductance curve of JFET.	**03**
	(b)	Explain transistor as a switch.	**04**
	(c)	Explain in detail different models used for AC analysis of BJT circuits.	**07**
		OR	
Q.4	**(a)**	Write applications of JFET.	**03**
	(b)	Explain the working of Emitter follower.	**04**

	(c)	Draw h parameter equivalent model for CE amplifier and derive the equation for input impedance, output impedance, Voltage gain and current gain for CE amplifier.	**07**
Q.5	**(a)**	Draw symbol, truth table and Boolean equation for EX-OR and EX-NOR gate.	**03**
	(b)	Compare different logic families.	**04**
	(c)	Explain the construction and principle of operation of Enhancement type P-channel MOSFET.	**07**

OR

Q.5	**(a)**	Design AND, OR and NOT gate using NAND Gate.	**03**
	(b)	Explain Transistor Transistor Logic (TTL).	**04**
	(c)	Explain the construction, working and drain characteristics of N-Channel JFET.	**07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I (NEW) EXAMINATION – WINTER 2023

Subject Code:3110016 Date:25-01-2024

Subject Name:Basic Electronics

Time:02:30 PM TO 05:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Draw symbol and truth table for AND, NOR, EX-OR logic gates.	**03**
	(b)	Discuss forward and reverse bias operation of a P-N junction diode with depletion region.	**04**
	(c)	Draw CB configuration and discuss its input and output characteristics with I_{CBO}, r_i, r_o, and current gain α.	**07**
Q.2	**(a)**	Describe Negative clamper circuit with necessary waveforms	**03**
	(b)	Describe RC filter with its limitations.	**04**
	(c)	Draw circuit diagram of Center-taped full wave rectifier and explain its operation with necessary waveforms and derivation of V_{dc}.	**07**
		OR	
	(c)	Describe diode approximations in detail.	**07**
Q.3	**(a)**	Draw circuit for Voltage doubler and explain its operation.	**03**
	(b)	Briefly explain the effect of coupling capacitor and bypass capacitor on small signal transistor amplifier.	**04**
	(c)	Draw circuit for Emitter feedback bias and explain in detail with its advantages and disadvantages.	**07**
		OR	
Q.3	**(a)**	State the difference between Avalanche breakdown and Zener breakdown.	**03**
	(b)	Discuss Zener diode as voltage regulator.	**04**
	(c)	Discuss load line, Q point, DC & AC load lines for BJT.	**07**
Q.4	**(a)**	Derive relation between α & β for BJT.	**03**
	(b)	Compare different BJT configurations.	**04**
	(c)	Describe working principle and applications of PIN Photo diode and Solar cell.	**07**
		OR	
Q.4	**(a)**	Briefly explain regions of operation for BJT.	**03**
	(b)	Discuss seven segment displays with its types.	**04**
	(c)	Draw CE configuration and discuss its input and output characteristics with I_{CEO}, r_i, r_o, and current gain β.	**07**

Q.5	**(a)**	Compare FET with BJT.	**03**
	(b)	Explain FET as a switch.	**04**
	(c)	State different logic families and compare them in terms of fan in, fan out, propagation delay, noise margin, power dissipation.	**07**

OR

Q.5	**(a)**	Construct AND & OR gates with diodes.	**03**
	(b)	Explain FET as an amplifier.	**04**
	(c)	Explain Drain characteristics and Transfer characteristics of JFET in detail with all related terms as Transconductance g_m, drain resistance r_d, and amplification factor μ.	**07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110016 Date:06-07-2024

Subject Name:Basic Electronics

Time:02:30 PM TO 05:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1	**(a)**	Explain series positive clipper with diagram.	**03**
	(b)	Draw and explain V-I characteristic of P-N junction diode.	**04**
	(c)	Draw and Explain bridge rectifier. Explain advantage and disadvantage of bridge rectifier over full wave rectifier.	**07**
Q.2	**(a)**	Drive the relation between current gain α and β for CE configuration.	**03**
	(b)	Draw the symbol of NPN & PNP transistor. Also state the advantage of transistor.	**04**
	(c)	Draw and explain input and output characteristic of transistor in CE configuration.	**07**
		OR	
	(c)	Compare CE, CB, and CC configuration with respect to different transistor characteristics.	**07**
Q.3	**(a)**	Explain V-I characteristics of tunnel diode.	**03**
	(b)	Explain Schottky diode in details.	**04**
	(c)	Comparison between P-N junction Diode and Zener Diode.	**07**
		OR	
Q.3	**(a)**	Advantage, disadvantage and application of LED.	**03**
	(b)	Compare PIN diode and Photo Diode.	**04**
	(c)	State the applications of Rectifier, and Comparison of Halfwave, Full-wave center- tap and Full-wave Bridge rectifier.	**07**
Q.4	**(a)**	List out the salient feature of emitter follower.	**03**
	(b)	Classification of Logic families in details.	**04**
	(c)	Explain application of Transistor as a switch.	**07**
		OR	
Q.4	**(a)**	Explain Negative Clamping circuit with diagram.	**03**
	(b)	Explain why NAND and NOR gate are called universal gate.	**04**
	(c)	Discuss MOSFET in details.	**07**
Q.5	**(a)**	Define the use of Coupling capacitor.	**03**
	(b)	Give Comparison of BJT and FET.	**04**
	(c)	Draw symbol and explain all logic gates in details.	**07**
		OR	
Q.5	**(a)**	Explain the properties and application of common base amplifier.	**03**
	(b)	State advantage, disadvantage and application of FET.	**04**
	(c)	Give comparison between different types of digital logic families.	**07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II EXAMINATION – WINTER 2024

Subject Code:3110016 **Date:17-01-2025**

Subject Name:Basic Electronics

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Sketch the characteristics of ideal diode and approximate characteristics of practical diodes. Briefly explain it.	**03**
	(b)	Draw two-diode full-wave rectifier circuit and explain its operation.	**04**
	(c)	Explain positive and negative voltage clamper circuit with waveforms.	**07**
Q.2	**(a)**	What are three modes of transistor operation, explain it.	**03**
	(b)	What is you understand by transistor biasing and why it is required.	**04**
	(c)	Compare CB, CE and CC configuration.	**07**
		OR	
	(c)	Explain the operation of voltage divider bias circuit using an npn transistor and write its voltage and current equation.	**07**
Q.3	**(a)**	What is photo-diode and how it works?	**03**
	(b)	What is varactor diode, and explain how it works?	**04**
	(c)	In CE amplifier has h_{ie} = 2.1 KΩ, h_{fe} = 75 and h_{oe} =1 μS, voltage divider resistance R_1= 68 kΩ and R_2= 56 kΩ, Rc=3.9 kΩ, R_E= 4.7 kΩ and R_L= 82 kΩ. Calculate input impedance, output impedance and voltage gain.	**07**
		OR	
Q.3	**(a)**	What is tunnel diode, and explain how it works?	**03**
	(b)	Explain how Zener diode maintains constant voltage across load with circuit.	**04**
	(c)	Draw single stage CE amplifier and analyze and explain how it works.	**07**
Q.4	**(a)**	Define saturation current, pinch off voltage and transconductance in JFET.	**03**
	(b)	Do comparison between BJT and FET and write it.	**04**
	(c)	Analyze and explain CB amplifier and find its input impedance, output impedance and voltage gain	**07**
		OR	
Q.4	**(a)**	Define performance parameter of JFET a.c. drain resistance, trnasconductance and amplification factor.	**03**
	(b)	In n-channel JFET with $V_{GS(off)}$ =-6 V and I_{DSS} = 3 mA Solve I_D value for V_{GS} =-1,-3, and -5 V.	**04**
	(c)	Consider a CB amplifier utilizing a BJT biased at Ic= 1 mA with Rc = 5 kΩ, R_L= 5 kΩ. Determine Rin, Avo, Av, and Ro.	**07**
Q.5	**(a)**	What is logic gates and state the rule used for OR and AND gate.	**03**
	(b)	Why NAND and NOR gates are called universal gate and draw truth table of NAND gate.	**04**
	(c)	Write and explain application of FET as amplifier and as a switch.	**07**

OR

Q.5 **(a)** Draw the logic circuit for the Boolean expression $Y = ABC + \overline{D}$. **03**

(b) Explain Transistor Transistor Logic (TTL). **04**

(c) Using common source circuit have a R_1=5.6 MΩ, R_2= 1 MΩ, R_D=R_S = 2.7 KΩ, rd = 100 KΩ and Y_{fs} = 3000 μS calculate input impedance, output impedance and voltage gain. **07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I&II EXAMINATION – SUMMER 2025

Subject Code:3110016 **Date:17-06-2025**

Subject Name:Basic Electronics

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Draw and explain constructional details and symbols for BJT.	**03**
	(b)	Draw and explain the V/I characteristics of a practical diode.	**04**
	(c)	Draw Common Base configuration and discuss its input and output characteristics with I_{CBO}, r_i, r_o, and current gain α.	**07**
Q.2	**(a)**	Define: clipper, Clamper, PIV of a Diode.	**03**
	(b)	Compare different rectifier circuits.	**04**
	(c)	Discuss diode approximations.	**07**
		OR	
	(c)	Explain full wave center-tape rectifier in detail with waveforms and derivation of average output voltage.	**07**
Q.3	**(a)**	Draw circuit diagram of positive clamper and explain the operation.	**03**
	(b)	Draw and explain circuit of AND gate using diodes.	**04**
	(c)	Explain thermal stability of BJT with stability factors S, S', S''.	**07**
		OR	
Q.3	**(a)**	Draw Hybrid equivalent model of transistor for CB, CE, CC configurations.	**03**
	(b)	Describe LED with its applications.	**04**
	(c)	Discuss operation and applications of PIN Photo diode and Solar cell.	**07**
Q.4	**(a)**	State the difference between Avalanche breakdown and Zener breakdown.	**03**
	(b)	Discuss seven segment display with its types.	**04**
	(c)	Explain biased positive and negative clipper circuits in detail.	**07**
		OR	
Q.4	**(a)**	For a BJT, α_{dc} is 0.98 and base current is 50 μA, then find β_{dc} , I_C & I_E.	**03**
	(b)	Compare CB, CE, CC configurations.	**04**
	(c)	Draw Emitter follower circuit and discuss its input and output characteristics.	**07**
Q.5	**(a)**	Compare BJT & FET.	**03**
	(b)	Explain OR gate using diodes.	**04**
	(c)	Describe MOSFET operation and draw the structures for D-MOSFET & E-MOSFET.	**07**
		OR	
Q.5	**(a)**	Discuss FET as an amplifier.	**03**
	(b)	Compare different logic families.	**04**
	(c)	Explain Drain characteristics and Transfer characteristics of JFET in detail with all related terms as Transconductance, drain resistance and amplification factor.	**07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II EXAMINATION – WINTER 2025

Subject Code:3110016 **Date:19-01-2026**

Subject Name:Basic Electronics

Time:02:30 PM TO 05:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Draw the vi characteristics for ideal diode and real diode.	**03**
	(b)	Explain the forward and reverse bias of PN junction.	**04**
	(c)	Discuss the operation of full bridge rectifier circuits with voltage and current waveforms.	**07**
Q.2	**(a)**	Draw the circuits for clipping the voltage waveforms.	**03**
	(b)	Explain the characteristics of Zener diode.	**04**
	(c)	Write short not on bipolar junction transistor (BJT).	**07**
		OR	
	(c)	Explain the operation of transistor as amplifier in Common Emitter(CE) configuration.	**07**
Q.3	**(a)**	Write key factors to achieve faithful amplifications.	**03**
	(b)	Discuss the importances of operating points in transistor.	**04**
	(c)	Discuss the load line analysis of transistor.	**07**
		OR	
Q.3	**(a)**	What is a stability factor?	**03**
	(b)	Explain the voltage divider biasing.	**04**
	(c)	Discuss the construction and operation of Schottky diode.	**07**
Q.4	**(a)**	What are the roles of coupling capacitor and bypass capacitor in amplifier circuits?	**03**
	(b)	Discuss the operation of common collector circuits analysis.	**04**
	(c)	Shows parametric comparisons between of CE, CB and CC.	**07**
		OR	
Q.4	**(a)**	Draw the circuits of seven segment LED display.	**03**
	(b)	What are the differences between FET and BJT?	**04**
	(c)	Write short note on MOSFET.	**07**
Q.5	**(a)**	Prepare the truth table for three basic logic gates.	**03**
	(b)	Explain the working principle of a field effect transistor (FET).	**04**
	(c)	Discuss the resistor transistor logic (RTL)circuits for basic gates.	**07**
		OR	
Q.5	**(a)**	Draw the circuit diagram of OR gate with diodes.	**03**
	(b)	Show the implementation of OR gate logic with use of NAND gate.	**04**
	(c)	Discuss the applications of EX-OR and EX-NOR gate with truth tables.	**07**